Theory of Computation

Closure

Lecture 36

Gaurav Raj

1.) $L_1 = \phi$

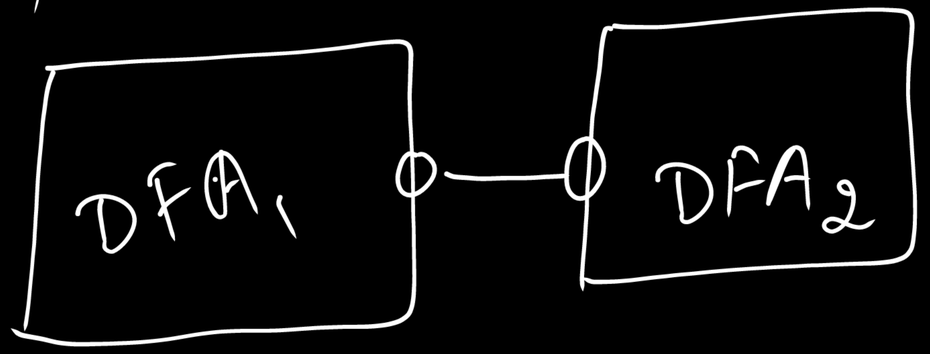
$L_2 = Reg$

Concatenation.

$L_1 \cdot L_2 = \phi$

$L_2 \cdot L_1 = \phi$

$\underbrace{Reg_2 \cdot Reg_1 = Reg}_{\text{Closed}}$



2.) $L_1 = a^*$

$L_2 = (a+b)^*$

$L_1 \cdot L_2 = (a+b)^*$

$\underline{L_2} \cdot \underline{L_1} = (a+b)^*$

$\cancel{a^* \cdot (a+b)^*}$

$(a+b)^* \cdot a^* = (a+b)^*$

3) $L_1 = a^+$

$L_2 = (a+b)^*$

$L_1 \cdot L_2 \equiv L_2 \cdot L_1$ True/false

$\underbrace{a^+(a+b)^*} = a(a+b)^*$

$\{a, aa, aaa\} \cdot \{\varepsilon + \ldots\}$

4) $L_1 = (a+b)$

$L_2 = (b+a)^+ \rightarrow \{a, b\}\{a, b, \ldots\}$

$L_1 \cdot L_2 = \cancel{(a+b)^+} = (a+b)^2 (a+b)^*$

$(a+b)(a+b)^+$

$L_2 \cdot L_1 = \cancel{(a+b)^+}$

Reverse

$L = a\Sigma^*$

$L^{Rev} = \Sigma^* \cdot a$

$L = a\Sigma^* b$

$L^{Rev} = b\Sigma^* a$

$L = \Sigma^* abb\Sigma^*$

$L^{Rev} = \Sigma^* bba\Sigma^*$

$(L^{Rev})^{Rev} = L$

$*$, $+$: Kleene $\rightarrow$ Closed.

$L = \text{Reg.}$

Subset of L = <u>may or may not</u> be Reg.

$L = a^* b^*$

$L_2 = a^n b^n$

$\cancel{a^*}$

$L = a^n$

$L_2 = \cancel{a^n b^n} : n \geq 0$

$$\begin{cases} L_1 = a^n \\ L_2 = (a^n + b^n) \end{cases}$$

a^n

$a^n b^n$

$a^n b^m : \quad n \geq 0, \; m \geq 0$

Thank You